PHS Hail Mary Orientation

Correlation and Linear Regression

Tabor Hoatson

Population Health Sciences Teaching Team
Harvard T.H. Chan School of Public Health

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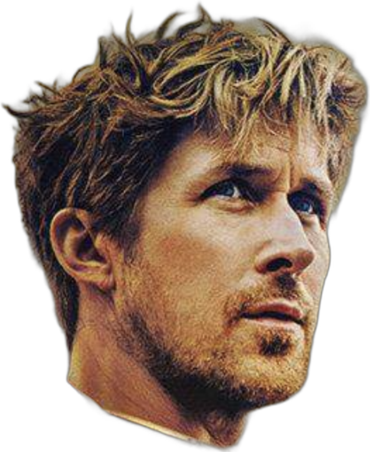


Introduction(s)

Me!

Hello!

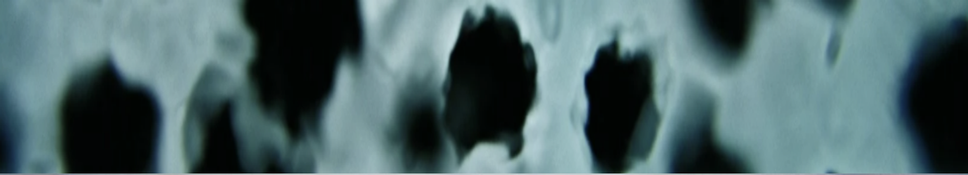
- As my title slide said, my name is Tabor (pronounced like “labor” union... shoutout HGSU ❤).
- I use he/him pronouns and I am a second year PHS PhD student in Epidemiology.
- I study psychiatric epi with a focus on stigma, suicide, health equity, and the biological impacts and origins of psychiatric adversity.
- If you want to know more or chat about any aspect of the PHS program, reach out! You can find me at hoatsont@g.harvard.edu.

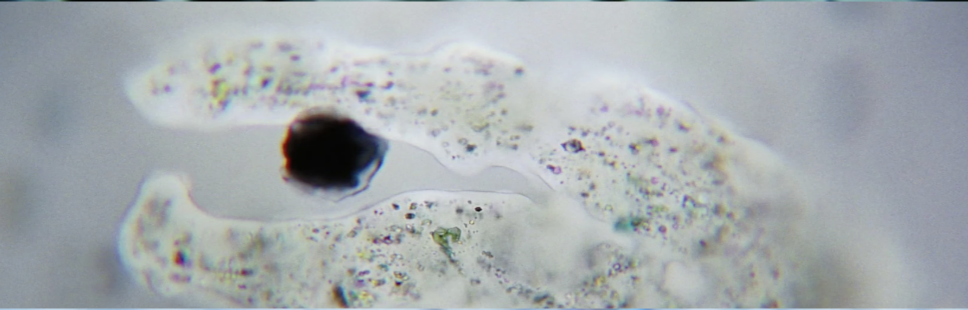
Content

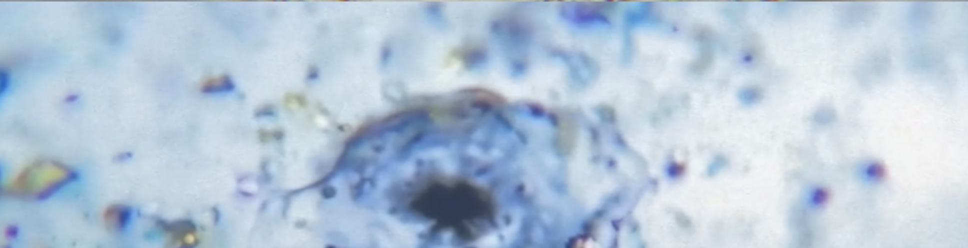
Today, we'll be building on Dylan and Ashlynn's lessons on math, variance, and covariance by discussing correlation and linear regression.

I'm hoping the lessons are feeling well-paced and accessible, but PLEASE interrupt me if I'm not making sense or (especially) you disagree with what I'm saying!

Correlation







Motivating example

Astrophage are microbial star-eaters which threaten death upon any solar system they infest.

- Grace and Rocky have discovered an Astrophage-eating predator endogenous to the Tau Ceti solar system. They name it, with great inspiration, Taumoeba.
- The intrepid duo watch Taumoeba eat Astrophage and reason that since Tau Ceti is the only star in their cluster without mass-extinction level dimming, Taumoeba must prevent Atrophage-induced star death.

But...

Motivating example

...What if a gradient of Taumoeba population density exists across solar systems, and Grace and Rocky just so happened to find one with an optimal level?

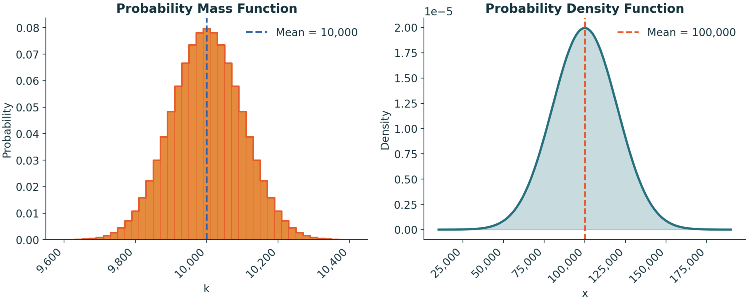
...How can we begin to explore the relation between Taumoeba presence in a solar system and Astrophage population density?

...These are questions to which Earth needs answers!!!!!!

...Looks like a job for Harvard's 2026 PHS PhD G1's

Quantifying the problem

- X is the number of Taumoeba per planetary surface area in a solar system
- Y is the number of Astrophage per star surface area in a solar system
- The numbers of Taumoeba and Astrophage in each solar system will be discrete random variables (count), but the average number of these creatures per planet/star surface area will be continuous!



Note: For a discrete random variable, $E[X] = \Sigma_x P(X = x)x$

Approaching the problem with correlation

Correlation

$$\rho_{xy} = \frac{\displaystyle\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\displaystyle\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\displaystyle\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

The three components in this formula have names.

Can anyone tell me what they are?

Approaching the problem with correlation

Correlation

$$\rho_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

- Covariance (numerator)
- Standard deviation of x (denominator)
- Standard deviation of y (denominator)

Completing the problem

Table 1: Local stellar neighborhood populations per habitat surface area

Solar System	Taumoeba	Astrophage
Sol	1	1,000,000
40 Eridani	2	1,500,000
Epsilon Eridani	1,400	800,000
Sirius	7,000	700,000
WISE 0855-0714	2,000	900,000
Tau Ceti	400,000	1,500
...	$\bar{x} = 68,400.5$	$\bar{y} = 816,916.67$

Completing the problem

Table 2: Applying components of the correlation formula to Table 1

System	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
Sol	−68,399.5	+183,083.3	4.68e9	3.35e10	−1.25e10
40 Eridani	−68,398.5	+683,083.3	4.68e9	4.67e11	−4.67e10
Epsilon Eridani	−67,000.5	−16,916.7	4.49e9	2.86e8	+1.13e9
Sirius	−61,400.5	−116,916.7	3.77e9	1.37e10	+7.18e9
WISE 0855-0714	−66,400.5	+83,083.3	4.41e9	6.90e9	−5.52e9
Tau Ceti	+331,599.5	−815,416.7	1.10e11	6.65e11	−2.70e11
		Sum	1.320e11	1.186e12	−3.268e11

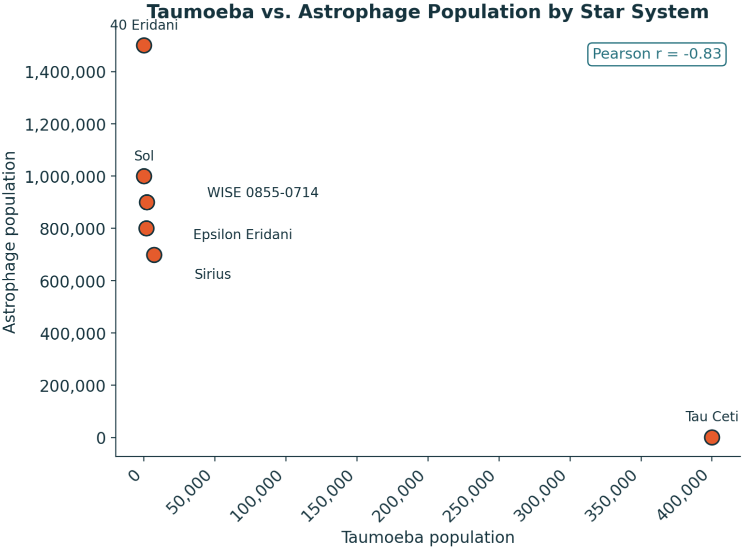
Completing the problem

$$\rho_{xy} = \frac{-3.268 \times 10^{11}}{\sqrt{1.320 \times 10^{11}} \times \sqrt{1.186 \times 10^{12}}} = \frac{-3.268 \times 10^{11}}{363{,}295 \times 1{,}088{,}984}$$

Result

$$\rho_{xy} \approx -0.826$$

Completing the problem



Correlation versus covariance

We saw in the formula slide that correlation is just covariance standardized into a unitless quantity.

What advantages might this have?

Correlation versus covariance

Correlation advantages

1. Scale free, resistant to unit changes
2. Comparability
3. Bounded range [-1, 1]
4. Consistent relationship between strength and size

Testing intuition

1. Which is stronger, a correlation of -0.8 or of 0.8?
2. What is the correlation of a random variable with itself?
3. What is the correlation between two independent random variables?
4. If two variables have covariance and correlation equal to zero, are they always independent?

Testing intuition

Answers

1. They're equal! Magnitude, not sign, determines correlation strength.
2. $\rho_{x,x}$ always equals 1!
3. 0! Independence guarantees the correlation numerator equals 0.
4. No! Correlation captures linear relationships only.

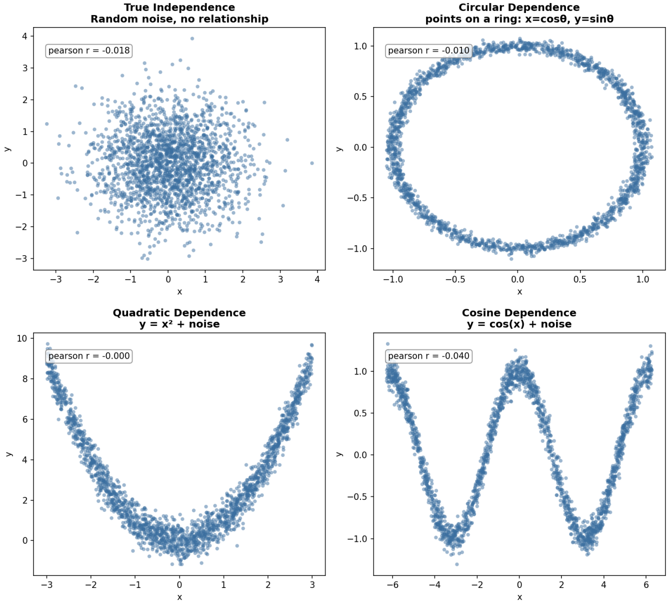
The motto...

Correlation $\neq$ causation!

But...

Causation also does not always imply correlation!

Non-linear dependence and Pearson's ρ



Linear Regression

Beyond Correlation

- Linear regression extends beyond correlation by quantifying how much one variable changes with each additional increase/decrease in another variable
  - For each additional Taumoeba per unit of planetary surface area, how many fewer Astrophage will exist per unit of solar surface area?
- Correlation and covariance do **not** provide an answer to such questions!
- Linear regression models can consider these questions **AND** can be extended to include **multiple relevant covariates!**

Mathematical Notation

For each additional Taumoeba per unit of planetary surface area, how many fewer Astrophage will exist per unit of solar surface area?

- $E[Y|X = (x + 1)] - E[Y|X = x]$

How might we model (make assumptions about a functional form) these expected value statements?

We can make the modeling assumption that the relation between X and Y can be drawn using a straight line!

- $y = mx + b \leftrightarrow y = b + mx$
- $E[Y|X = x] = \beta_0 + \beta_1 X$

b, β_0 are intercept parameters. m, β_1 are slope parameters.

Estimands and Estimates

$$E[Y|X = (x + 1)] - E[Y|X = x] = \beta_0 + \beta(x + 1) - \beta_0 - \beta_1 x = \beta_1$$

β_1 is our estimate of the estimand of interest!

Estimators and Estimation

Parameters from a generalized linear model are frequently estimated using ordinary least squares (OLS) or maximum likelihood estimation (MLE) estimators. We'll get to this in future lessons this semester!

- The OLS solution for the most likely β_1 parameter given observed data is:

$$\hat{\beta}_1 = \frac{\Sigma_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\Sigma_{i=1}^{n}(x_i - \bar{x})^2}$$

* How the $\hat{\beta}_1$ formula relates to correlation:

1. Correlation formula:

$$\rho = \frac{Cov(X,Y)}{\sqrt{Var(X)} \cdot \sqrt{Var(Y)}}$$

2. multiply by $\sqrt{var(Y)}$:

$$\rho \cdot \sqrt{var(Y)} = \frac{Cov(X,Y)}{\sqrt{var(X)} \cdot \sqrt{\cancel{var(Y)}}} \cdot \sqrt{\cancel{var(Y)}}$$

3. Divide by $\sqrt{var(X)}$:

$$\rho \cdot \frac{\sqrt{var(Y)}}{\sqrt{var(X)}} = \frac{Cov(X,Y)}{\sqrt{var(X)}} \Big/ \sqrt{var(X)}$$

4. Reaching the $\hat{\beta}_1$ formula from the previous slide:

$$\rho \cdot \underbrace{\frac{\sigma_Y}{\sigma_X}}_{\text{standard deviation}} = \boxed{\frac{Cov(X,Y)}{Var(X)}} = \frac{\text{covariance}(X,Y)}{\text{variance}(X)}$$

Questions and Acknowledgements

Questions?

Acknowledgments

HUGE thank you to the previous Teaching Fellows who contributed to these slides!

- Leah Martin
- Alex John Zapf
- Shu Saha
- Matt Lee
- Louisa Smith
- Chris Boyer

And... to the creator of the metropolis beamer theme: Matthias Vogelgesang!